2D: ECUACIONES DE MOMENTO 2D:

$$\begin{cases} \rho \dfrac{Du_1}{Dt} - \nu(\Delta u_1) + \dfrac{\partial p}{\partial x} = f_1(x,y,t) \\ \rho \dfrac{Du_2}{Dt} - \nu(\Delta u_2) + \dfrac{\partial p}{\partial y} = f_2(x,y,t) \end{cases}$$

(si añadimos conservac. masa se denom. Ec. Navier-Stokes)

de forma más desarrollada en derivadas parciales:

$$\begin{cases} \rho\left[\dfrac{\partial u_1}{\partial t} + u_1 \dfrac{\partial u_1}{\partial x_1} + u_2 \dfrac{\partial u_1}{\partial x_2}\right] - \nu\left(\dfrac{\partial^2 u_1}{\partial x_1^2} + \dfrac{\partial^2 u_1}{\partial x_2^2}\right) + \dfrac{\partial p}{\partial x} = f_1 \\ \rho\left[\dfrac{\partial u_2}{\partial t} + u_1 \dfrac{\partial u_2}{\partial x_1} + u_2 \dfrac{\partial u_2}{\partial x_2}\right] - \nu\left(\dfrac{\partial^2 u_2}{\partial x_1^2} + \dfrac{\partial^2 u_2}{\partial x_2^2}\right) + \dfrac{\partial p}{\partial y} = f_2 \end{cases}$$

en forma más compacta:

$$\begin{cases} \rho\left[\dfrac{\partial u_1}{\partial t} + \vec{u} \cdot \vec{\text{grad}}\, u_1\right. - \nu\left(\text{Div}(\vec{\text{grad}}\, u_1)\right) + \dfrac{\partial p}{\partial x} = f_1 \\ \rho\left[\dfrac{\partial u_2}{\partial t} + \vec{u} \cdot \vec{\text{grad}}\, u_2\right. - \nu\left(\text{Div}(\vec{\text{grad}}\, u_2)\right) + \dfrac{\partial p}{\partial y} = f_2 \end{cases}$$

VECTOR ACELERACIÓN DE UN FLUIDO

3D
$$\vec{u}(u_1, u_2, u_3) \quad , \quad u_1(x_1, x_2, x_3, t)$$
$$du_1 = \frac{\partial u_1}{\partial t} dt + \frac{\partial u_1}{\partial x_1} dx_1 + \frac{\partial u_1}{\partial x_2} dx_2 + \frac{\partial u_1}{\partial x_3} dx_3$$

2-D
$$du_1 = \frac{\partial u_1}{\partial t} dt + \frac{\partial u_1}{\partial x_1} dx_1 + \frac{\partial u_1}{\partial x_2} dx_2$$

por definic. de velocidad
$$\left\{ \begin{array}{l} dx_1 = u_1 dt \\ dx_2 = u_2 dt \end{array} \right\} \rightarrow \frac{du_1}{dt} = \frac{\partial u_1}{\partial t} + \frac{\partial u_1}{\partial x_1} u_1 + \frac{\partial u_1}{\partial x_2} u_2$$

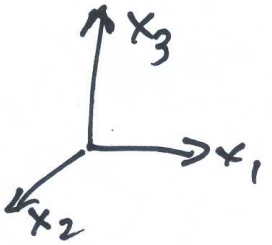
$$a_{x_1} = \frac{du_1}{dt} = \frac{Du_1}{Dt} = \frac{\partial u_1}{\partial t} + u_1 \frac{\partial u_1}{\partial x_1} + u_2 \frac{\partial u_1}{\partial x_2}$$

$$a_{x_2} = \frac{du_2}{dt} = \frac{Du_2}{Dt} = \frac{\partial u_2}{\partial t} + u_1 \frac{\partial u_2}{\partial x_1} + u_2 \frac{\partial u_2}{\partial x_2}$$

$$\vec{a} = \begin{bmatrix} a_{x_1} \\ a_{x_2} \end{bmatrix} = \begin{bmatrix} \frac{\partial u_1}{\partial t} + u_1 \frac{\partial u_1}{\partial x_1} + u_2 \frac{\partial u_1}{\partial x_2} \\ \frac{\partial u_2}{\partial t} + u_1 \frac{\partial u_2}{\partial x_1} + u_2 \frac{\partial u_2}{\partial x_2} \end{bmatrix}$$

$$\frac{D\vec{u}}{Dt} = \boxed{\frac{\partial \vec{u}}{\partial t} + \vec{u} \vec{\nabla} \vec{u}}$$

"nomenclatura"



$$= a_{x_1} \vec{i} + a_{x_2} \vec{j}$$

"Campo vectorial de aceleración $\vec{a}$"

$$\left\{ \begin{array}{l} a_{x_1} = \frac{\partial u_1}{\partial t} + \vec{u} \cdot \vec{\text{grad}}\, u_1 \\ a_{x_2} = \frac{\partial u_2}{\partial t} + \vec{u} \cdot \vec{\text{grad}}\, u_2 \end{array} \right\}$$